

Rating Model Safety in Eighty Forward Passes

Abstract

Deciding whether an open-weight checkpoint is safety-aligned normally costs hundreds of harmful generations plus a judge model, per checkpoint. Cheap activation-based alternatives exist, but every one we could locate in primary text is validated on four or fewer architecture families, and none resamples on the weight lineage — the unit at which checkpoints derived from a common base are not independent. We ask whether any member of this class predicts behaviour once the panel is large enough to say. On 52 checkpoints over 28 weight lineages and 11 families, the first-decoding-step logit-gap margin — 80 forward passes, no generation, no judge, no benchmark run, no reference model — predicts the judged plain-harmful refusal rate at Spearman $\rho = 0.694$ [0.495, 0.822] with the lineage as the resampling unit, and 0.564 [0.140, 0.826] with members aggregated within lineage. It transfers across provenance blocks (0.667 on 19 archived members versus 0.668 on 33 new ones), is not a parameter-count proxy (partial $\rho = 0.676$), and beats our reimplementations of the closest published activation scanner by +0.421 [0.169, 0.684]. The margin is read on harmful prompts, so it is not harmful-prompt-free; the benign variant collapses to 0.129. The same panel retracts two of our own earlier results: a paraphrase refit of that scanner falls from +0.296 at 7 lineages to +0.099 [−0.027, 0.244] at 28, localised to the block that produced it, and a claimed read-versus-act coupling of 0.629 is 89.6% a between-axis-type contrast by exact decomposition, its within-axis value being 0.547 with an interval covering zero. The difference between a cheap safety score that works and one that does not is invisible at seven lineages and legible at twenty-eight.

1 Introduction

Anyone who downloads an open-weight checkpoint faces a question with no cheap answer: is this model safety-aligned, and how much? The standard answer is a harmful-prompt benchmark such as AdvBench [Zou et al., 2023b], JailbreakBench [Chao et al., 2024] or HarmBench [Mazeika et al., 2024], several hundred generations scored by a judge model [Zheng et al., 2023], and a repeat of the whole procedure for every attack template of interest. The evaluator must hold, transmit and store harmful content, must pay for a judge, and must trust that the checkpoint was not tuned to refuse exactly the items it will be shown.

The stakes are set by scale. Hugging Face hosts hundreds of thousands of derived checkpoints, a growing fraction of them explicitly *uncensored* community fine-tunes, and the cheapest of these is produced by a weight edit — *abliteration* — that orthogonalizes every write against a single refusal direction [Arditi et al., 2024]. A platform, a downstream deployer or a regulator wanting to triage such a population needs a score that costs seconds per model.

The published attempts at such a score are validated on panels far too small to support them, and the panel size is the part nobody reports. AMS [Messenger, 2026] scans activation geometry over 14 configurations from 4 architecture families and reports Pearson $r = -0.546$ ($p = 0.043$) against behavioural compliance — its directly comparable rank statistic is $\rho = -0.423$ at $p = 0.13$, which is not significant. RAS/SafeVec [Huang et al., 2026] needs unsafe prompts, jailbreak prompts

and a safety-aligned reference model, and reports no correlation coefficient, no n and no resampling unit. VISAGE [Peng et al., 2024] evaluates a harmful benchmark at every weight perturbation. AQI [Borah et al., 2025] claims correlation with external judges with no locatable n or coefficient. The one audit that exceeds 20 checkpoints — 273 of them — scores a *binary* uncensored label and presumes an attested reference model plus a full weight download [Hurtado, 2026]. Of the five model-level internal safety scores we could locate in primary text, five validate at four or fewer architecture families, four at fourteen or fewer checkpoints, and **zero** resample on the weight lineage¹. Whether any member of this class predicts behaviour once the panel is large enough to say has, until now, not been tested by anyone.

This paper tests it, and one score passes. On a frozen panel of 52 checkpoints over 28 weight lineages and 11 architecture families, the first-decoding-step logit-gap margin read on harmful prompts — 80 forward passes, zero generations, zero judge calls, zero benchmark runs, zero reference models per checkpoint — predicts the judged plain-harmful refusal rate at Spearman $\rho = 0.694$ [0.495, 0.822] with the lineage as the resampling unit, and 0.564 [0.140, 0.826] when members are first aggregated within lineage. It is not a small-panel artifact: split by provenance, the 19 archived members give $\rho = 0.667$ and the 33 newly measured ones $\rho = 0.668$, a difference of -0.0004 . It survives a partial correlation on $\log_{10}$ parameter count (0.676), 28 leave-one-lineage-out folds ([0.661, 0.726]) and 11 leave-one-family-out folds ([0.650, 0.772]) without a sign change, and it beats our reimplementations of the closest published activation scanner by $+0.421$ [0.169, 0.684] on the same resampled lineages.

The honesty statement that must travel with that number, and that we put in the abstract rather than a footnote, is what the score does *not* remove. It reads the margin on harmful prompts, so it is not harmful-prompt-free; the benign-prompt variant collapses to $\rho = 0.129$ $[-0.168, 0.436]$. What it removes is everything downstream of the forward pass: no generation, no judge, no benchmark, no reference model.

The same panel is what retires the rest of this project’s programme, and we report those retractions as results rather than as limitations. Two iterations ago we proposed an act-side score — α_{50} , the steering coefficient at which a fresh generation on a benign prompt refuses half the time — and it failed. The five-check falsification battery built to explain that failure could not rank cheap scores either. One positive lead survived, a paraphrase refit of AMS’s contrast set that lifted criterion validity from $\rho = 0.358$ to 0.654 on 7 lineages; measured on the same 28-lineage panel it collapses to $\Delta_A = +0.099$ $[-0.027, 0.244]$, with the effect localised entirely to the original 19-member block. Two scores, one panel, opposite outcomes: that contrast is what makes the panel informative rather than merely large.

The previous draft of this paper additionally claimed that reading and steering along one refusal axis are strongly coupled, at $\rho = 0.629$ over 70 (member, axis) pairs. A reviewer identified that statistic as dominated by a between-axis-type contrast, and re-analysis confirms it: an exact two-way decomposition attributes 0.896 of the pooled coefficient to variation between axis types². The correct primary statistic — within the canonical axis, across models — is $\rho = 0.547$ with a lineage-clustered interval of $[-0.031, 0.930]$ that covers zero. We report the weaker claim.

¹Code: <https://github.com/AMGrobelsnik/ai-invention-c9546e-rating-model-safety-in-eighty-forward-pa/tree/main/round-5/research-1>

²Code: <https://github.com/AMGrobelsnik/ai-invention-c9546e-rating-model-safety-in-eighty-forward-pa/tree/main/round-5/evaluation-1>

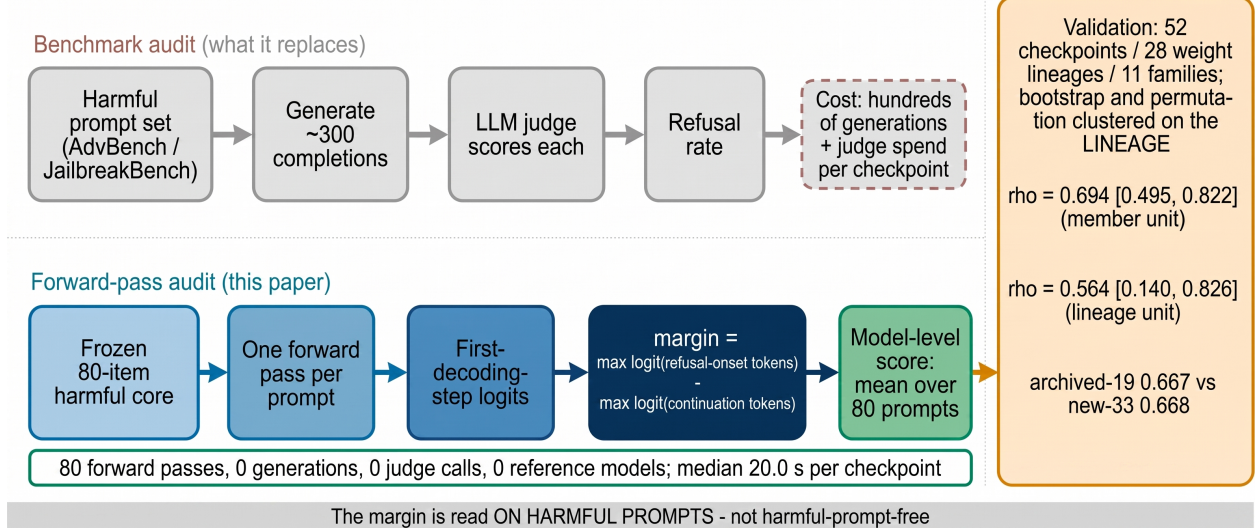


Figure 1: The audit this paper validates, against the audit it replaces. The benchmark path (top) requires holding harmful prompts, generating from them, and paying a judge, per checkpoint. The forward-pass path (bottom) reads the refusal-versus-affirmation logit margin at the first decoding step over a frozen 80-item harmful core: 80 forward passes, no generation, no judge, no benchmark run and no reference model. Validation is where the contribution lies: the resampling unit is the weight lineage, not the checkpoint.

1.1 Summary of Contributions

- **A cheap safety score that survives a fourfold panel increase** (§5.1)³. The first-decoding-step logit-gap harmful margin predicts judged plain-harmful refusal at $\rho = 0.694$ [0.495, 0.822] (member unit, lineage-clustered bootstrap) and 0.564 [0.140, 0.826] (lineage unit) over 52 members / 28 lineages / 11 families, with a Monte-Carlo lineage-permutation p at the 5×10^{-6} design floor. The archived-19 vs new-33 block split gives 0.667 vs 0.668. Cost: 80 forward passes, 0 generations, median 20.0 s per checkpoint including download.
- **The first model-level criterion validation in this class that resamples on the weight lineage** (§2, §5.1). Zero of five published model-level internal safety scores do; the widest published family axis is four, against our eleven. We claim neither the scalar (it is Li and Liu’s [Li and Liu, 2025]) nor the largest checkpoint panel (Hurtado’s [Hurtado, 2026]) — only the conjunction.
- **A companion negative that makes the positive interpretable** (§5.2)⁴. On the identical panel the AMS paraphrase refit gives $\Delta_A = +0.099$ [−0.027, 0.244] against an archived +0.296, an independently authored paraphrase set gives −0.152, and the effect is localised: the archived 19 reproduce +0.2963 to 2.6×10^{-4} while the 33 new members give −0.016. Panel size is not a nuisance parameter in this lane; it decides which of two same-cost scores you would ship.

³Code: <https://github.com/AMGrobelnik/ai-invention-c9546e-rating-model-safety-in-eighty-forward-pa/tree/main/round-5/experiment-1>

⁴Code: <https://github.com/AMGrobelnik/ai-invention-c9546e-rating-model-safety-in-eighty-forward-pa/tree/main/round-4/experiment-1>

- **The read-versus-act coupling, re-estimated without the axis-type confound (§5.3).** Within the canonical axis across 14 detection-powered members: $\rho = 0.547$ $[-0.031, 0.930]$, exhaustive permutation $p = 0.149$; lineage unit 0.821 $[0.348, 1.000]$. The pooled 0.629 is demoted to a secondary and decomposed: 0.896 between axis type, 0.036 between members, 0.069 residual. Verdict: `COUPLING_IS_AXIS_TYPE_CONTRAST` and `UNDERPOWERED`, both firing.
- **The verdict rule is n -asymmetric, and we quantify by how much (§5.3).** Simulating the study’s own bootstrap over 141 cells, `AT_CHANCE` is unreachable below 40 items per class ($P = 0.000$ at the pre-registered gate; Hanley–McNeil closed form $n = 65$; first attainable at $n = 80$), while `READS` fires with probability 1.000 under perfect separation at $n = 7$. The tally is now reported twice — 20/1/0/9 over all 30 members, 13/1/0/0 over the 14 powered ones — and the ablated-arm claim is re-carried on refusal *rates* (exhaustive permutation $p = 0.0026$; paired sign test 10/10, $p = 0.0020$), needing no AUROC at all.
- **The detection label and the axis share a lexical basis, and the consequence is measured (§5.3).** Re-labelling 660 stratified items with a five-class semantic rubric barely moves the pooled AUROC ($0.834 \rightarrow 0.821$, paired -0.013 $[-0.067, +0.030]$), but splits it: 0.897 $[0.864, 0.922]$ on canonically-worded refusals against 0.611 $[0.542, 0.686]$ on non-canonical ones, which does not clear the members’ own random-direction reading band. Verdict: `READS_CANONICAL_WORDING_ONLY`.
- **The one published leakage control, run on our own headline (§5.3).** Re-estimating every centring and scaling statistic inside the training fold under leave-one-prompt-out moves axis-A AUROC by -0.0205 $[-0.0352, -0.0071]$, against the -0.336 the same control produced on its author’s data [Mehta, 2026]; the control on the control moves random axis D by -0.0020 .
- **Three named measurement pathologies, each measured on our own published numbers (§6).** Item-pool provenance is leakage type [L3.3] in Kapoor and Narayanan’s taxonomy [Kapoor and Narayanan, 2023]; the aggregation unit moving ρ from 0.358 to 0.821 is ecological correlation in Robinson’s sense [Robinson, 1950]; the $7 \rightarrow 28$ lineage collapse is small-sample correlation instability, and Schönbrodt and Perugini’s own table puts our $n = 28$ six to nine times below the point of stability [Schönbrodt and Perugini, 2013, 2018]. We claim the instances, not the phenomena.

2 Related Work

Static, benchmark-free safety metrics, and how they are validated. AMS [Messenger, 2026] computes a standardized mean difference of projections onto a diff-in-means direction at 96 forward passes per model. RAS/SafeVec [Huang et al., 2026] extracts layer-wise refusal directions from a safety-aligned reference model and scores a target by hidden-state alignment. VISAGE [Peng et al., 2024] measures a weight-space safety basin, requiring a harmful benchmark at every perturbation. AQI [Borah et al., 2025] is a prompt-invariant latent-geometry diagnostic. RAS and VISAGE we do not run, for reasons fixed by a primary-source reimplement audit ⁵: every RAS-scored checkpoint is $\geq 4B$ and none overlaps any panel at our scale, and VISAGE at published fidelity costs 4,800 generations and roughly 28 hours per 1B model on CPU. AMS and Logit-Gap Steering [Li and Liu, 2025] we reimplement and run.

⁵Code: <https://github.com/AMGrobelsnik/ai-invention-c9546e-rating-model-safety-in-eighty-forward-pa/tree/main/round-1/research-1>

What is uniform across these papers is the thinness of the validation panel, and a dossier read all five in primary text to establish it. AMS validates at 14 configurations over 4 families; RAS states a qualitative correlation with attack success rate across 3 families with no coefficient, n or resampling unit; AQI’s abstract claims correlation with external judges with neither n nor coefficient locatable; Logit-Gap Steering validates a *per-prompt* margin against its own attack’s success rate across suffix strategies, with family-clustered inference at $n = 3$; Hurtado’s ablation audit [Hurtado, 2026] exceeds 20 checkpoints at 273 but against a binary label, with an attested reference and full weight access. Not one resamples on the weight lineage — the unit at which the statistical dependence actually lives, since one pretrained base and its derivatives are not independent observations. That is the gap §5.1 fills, and it is a gap in *validation practice*, not in scalar design.

The scalar we validate is not ours. Logit-Gap Steering [Li and Liu, 2025] defines the difference between the top refusal-token logit and the top affirmative-token logit at the first decoding step as “the per-prompt safety margin that alignment provides”, and reports that the aligned model’s gap exceeds the base model’s on 97.5–99.8% of toxic prompts, with position-1 accounting for 92–99% of the refusal decision depending on family. A full-text extraction confirms that no cross-model margin-versus-behaviour correlation exists anywhere in that paper: all 28 correlation matches are token-level, suffix-level, or cited from elsewhere. One estimand difference must be declared: their affirmative token is selected *per prompt* as the highest-logit one, making their gap an attack-relevant minimum, whereas our fixed-lexicon maximum is a different quantity. Xu and Sheng [Xu and Sheng, 2026] use refusal vectors as a provenance fingerprint over 76 offspring models, identifying the base family at 100% accuracy — model identity, not safety behaviour, which is the cleanest available contrast to what we predict.

Detection versus intervention. Galeone et al. [Galeone et al., 2026] establish that a detection direction at AUC 1.000 can sit at $\cos = 0.12$ from the direction that produces the behaviour, and propose a *functional* criterion: the steerable case is where the intervention direction also detects. Mehta [Mehta, 2026] is the closest published neighbour to the read-versus-act claim our previous draft led with, and it is a mirror image: one direction detects alignment faking at leave-one-query-out AUROC 0.87 on Llama-3.1-8B while steering over 2,000 runs “barely changes compliance” (Cohen’s $h = +0.057$, Fisher $p = 0.41$). Three distinctions were verified in full text ⁶: his dissociation is assembled *across two models* (the steering null is on Qwen3-32B, where his own detection falls to 0.43), his probe is a two-layer MLP rather than the steered unit vector, and no activation norm is reported, so his coefficient is not convertible to our contrast units. One concession is forced: he does steer a refusal axis and get a null. The most transferable thing in his paper is not the dissociation but the leakage control it survived — his own AUROC falls from 0.761 to 0.43 under per-fold residualisation and leave-one-query-out, and it reaches 0.63 on a control condition where the effect cannot exist. We run that control on our own headline in §5.3. Nadaf [Nadaf, 2026] independently reports that steering succeeds where the logit lens cannot decode across 4,032 concept-layer pairs while the converse is “nearly empty (3 of 72)”, which makes coupled read-act the expected case.

Activation scores on ablated checkpoints. Two incumbents bound what we may claim. Hurtado [Hurtado, 2026] separates 57 public ablations from 37 benign fine-tunes at AUROC

⁶Code: <https://github.com/AMGrobelsnik/ai-invention-c9546e-rating-model-safety-in-eighty-forward-pa/tree/main/round-4/research-1>

0.95, but the activation leg is a thresholded ratio (AUROC 0.84) that “certifies whether the refusal mechanism is present, not whether a model is harmless”, and it “presumes an attested reference”. Llorente-Saguer [Llorente-Saguer, 2026] already runs base / instruction-tuned / ablated Qwen triplets and reports that “both ablated variants achieve AUROC at most 0.015 below their instruction-tuned counterparts”, noting that its axis “is not the refusal direction itself, since it survives ablation”. Any claim to be first to read an activation safety score on ablated checkpoints is therefore withdrawn. What survives, and what §5.3 reports, is narrower and compatible: the *refusal axis specifically* goes quiet on ablated checkpoints, because the refusals it would read are gone.

Steering-vector reliability. Non-identifiability is established: steering vectors admit “large equivalence classes of behaviorally indistinguishable interventions” [Venkatesh and Kurupath, 2026]. Unreliability has geometric predictors across 36 datasets [Braun, 2026], and the safety cost of steering has been catalogued [Li et al., 2026]. Success is partly predictable *ex ante* at $\rho = +0.86$ to $+0.91$ across 24 concept families [Billa, 2026], though the Linear Accessibility Profile could not have predicted our axis comparison, because it never sees the steering direction and both of our axes score identically. Refusal is multi-directional: eleven category directions, several near-orthogonal, yield “nearly identical refusal to over-refusal trade-offs” [Joad et al., 2026], and category-specific directions can be composed for control [Alagharu et al., 2026]. Petrov [Petrov, 2026] was the top refutation risk for our axis comparison, reporting that changing only the contrast baseline “produces no functional refusal directions at any tested weight level on any tested layer” by “reducing the extracted direction magnitude below the threshold at which weight-matrix projection perturbs the residual stream”; §5.4 settles it on 30 checkpoints in axis-contrast units, which normalise the axis magnitude by construction. Wu et al. [Wu et al., 2026] show a “steerability emerges with scale” result dissolves under exactly that normalisation.

Auditing a safety measurement, and the pathologies we name. The battery framing is prior art in kind and we say so. Wang et al. [Wang et al., 2026] separate “construct validity . . . metric validity . . . criterion validity”, run pre-specified positive and negative controls, and survive “leave-one-organization-out and organization-clustered bootstrap”; they are also the source of the field-local warning that a small panel manufactures results — a correlation moving “from -0.64 at $n = 7$ to $+0.02$ at $n = 18$ ”, with “a quarter of random size-7 subsets” showing $|\rho| \geq 0.5$ despite a near-zero full-panel value. Weng et al. [Weng et al., 2026] operationalise rubric-semantics invariance under certified-equivalent rewrites and state the discrimination requirement outright. The methodological ancestor of both is the sanity-check literature for saliency maps [Adebayo et al., 2018]. Outside machine learning, the three phenomena §6 names have canonical statements: Kapoor and Narayanan’s leakage taxonomy [Kapoor and Narayanan, 2023], whose type [L3.3] *sampling bias in test distribution* is our item-pool case exactly; Robinson’s demonstration that an ecological correlation can differ in sign from its individual counterpart [Robinson, 1950], extended by Openshaw’s modifiable-area-unit literature [Openshaw, 1984]; and Schönbrodt and Perugini’s point-of-stability analysis [Schönbrodt and Perugini, 2013, 2018]. Simpson’s note [Simpson, 1951] names the paradox but its setting is categorical contingency tables, so Robinson is the closer analogue.

Refusal geometry and dynamics. Arditi et al. [Arditi et al., 2024] show refusal is mediated by a single direction and introduce the weight edit the ablation community built on; representation engineering [Zou et al., 2023a], activation addition [Turner et al., 2023] and contrastive activation

addition [Rimsky et al., 2023] supply the steering machinery. Qi et al. [Qi et al., 2024] show aligned and unaligned generative distributions differ mainly over the first few output tokens; Yin et al. [Yin et al., 2025] trace a probe refusal score across token positions, an observable we adopt rather than coin. Korznikov et al. [Korznikov et al., 2025] report random steering raising harmful *compliance* from 0% to 1–13% at an identically calibrated coefficient; §5.6 supplies the matching measurement for the direction they do not test. Our behavioural axes follow AdvBench [Zou et al., 2023b], JailbreakBench [Chao et al., 2024] and XSTest [Röttger et al., 2023], with judge scoring in the style of Zheng et al. [2023]; Hasan and Biswas [Hasan and Biswas, 2026] find over-refusal and harmful compliance nearly uncorrelated ($r = -0.032$, $p = 0.89$) across 21 open-weight models, which is why the three axes are predicted separately. The critical-slowness programme [Scheffer et al., 2009, 2012, Dakos et al., 2012, Bury, 2023] supplied the indicators for this project’s first iteration; that arm is closed and summarised in Appendix A.

3 Preliminaries

Panels and the resampling unit. Three panels appear, and every claim names the one it rests on. The *depth* panel is six Qwen3 [Yang et al., 2025] checkpoints (0.6B and $1.7\text{B} \times \text{base} / \text{instruct} / \text{abliterated}$) measured exhaustively ⁷. The *read-versus-act* panel is 30 checkpoints over 7 lineages, each measured in both roles of five axes ⁸. The *scale* panel is 52 analysed members over 28 weight lineages and 11 architecture families at $\leq 4.2\text{B}$, drawn from a frozen manifest of 137 verified checkpoints over 93 lineages ⁹, and split for diagnostic purposes into the 19 members archived from iteration 2 and the 33 measured afresh. The *lineage* — one pretrained base and everything derived from it — is the resampling and permutation unit for every model-level claim.

Aggregation unit. Because a lineage carries between one and four members, a model-level correlation can be computed two ways. *Member level* means one row per checkpoint with the bootstrap and the permutation clustered on the lineage label; *lineage level* means one row per lineage, each the arithmetic mean over that lineage’s defined members of both the score and the outcome. Both are legitimate; they are not interchangeable, and §5.5 measures how far apart they are. Every correlation below is labelled and Table 5 gives both for every score.

Prompt sets. All frozen and shipped: 40 vetted everyday harmless user turns over 10 topics; 594 deduplicated AdvBench/JailbreakBench harmful behaviours with an 80-item stratified core; 400 jailbreak items; XSTest’s 250 safe and 200 unsafe items; 256 harmful/benign contrast items reserved for layer selection only (exact overlap 0 with any evaluation set, maximum cosine 0.652 against a 0.85 threshold); 200 WikiText fluency passages.

The four scores. All are read from forward passes only. **Logit-gap harmful** is the difference between the maximum logit over a per-tokenizer-family empirical refusal-onset lexicon and the maximum over a continuation lexicon, at the first decoding step, averaged over the frozen 80-item harmful core — 80 forward passes. **Logit-gap benign** is the same quantity on 40 harmless

⁷Code: <https://github.com/AMGrobelsnik/ai-invention-c9546e-rating-model-safety-in-eighty-forward-pa/tree/main/round-2/experiment-1>

⁸Code: <https://github.com/AMGrobelsnik/ai-invention-c9546e-rating-model-safety-in-eighty-forward-pa/tree/main/round-4/experiment-2>

⁹Code: <https://github.com/AMGrobelsnik/ai-invention-c9546e-rating-model-safety-in-eighty-forward-pa/tree/main/round-1/dataset-1>

prompts. **Our-AMS** σ is our reimplementation of Messenger [2026] — $(\mu_+ - \mu_-)/\sigma_{\text{pooled}}$ on the diff-in-means direction, final prompt token, 40–80% relative-depth sweep, 96 forward passes. α_{50} , retained only as this project’s retired incumbent, is the steering coefficient at which a fresh benign-prompt generation refuses half the time; it costs 265 generations per member.

The refusal-onset lexicon is *empirical*, derived per tokenizer family from tokens that actually open refusals under greedy decoding, and it is never back-filled across families: five scale-panel members whose family has no lexicon entry receive NULL primary columns with reason MISSING_FAMILY_LEXICON, and a clearly labelled union-of-all-families secondary column is reported beside the null rather than substituted for it.

Steering axes. Five axes are fitted per model at the same site on the same benign prompts: **A**, the canonical response contrast between four hand-written canned refusals and four compliances; **B**, a token-disjoint paraphrase axis built from 24 refusal paraphrases containing zero frozen refusal openers; **C**, a norm-matched non-safety stylistic axis (formal versus casual); **D**, matched random unit directions; **E**, the harmful-versus-benign *prompt* contrast. The steering site is $L = \text{round}(0.25 n_{\text{layers}})$.

Axis-contrast units. The steering hook adds $\alpha \text{NORM}_L \hat{x}$ to the residual stream, where NORM_L is the median residual-stream norm at the steering layer. Because raw diff-in-means norms differ up to fourfold between axes, all dose comparisons are reported per *axis-contrast unit* $c = \alpha \text{NORM}_L / \|X\|$, which divides out the axis magnitude and is what settles the magnitude-collapse rival explanation [Petrov, 2026]. The conversion was verified exact (absolute error 0.0) against 459 archived grid cells.

The refusal criterion. A generation counts as a refusal when its opening matches an anchored refusal regex. Because the regex is lexical, §5.3 and §5.4 additionally re-score with a four-class semantic judge and with a five-class rubric carrying an explicit non-canonical-refusal class, and every semantic rate is reported against a control false-positive floor measured on the same filtered population.

A tokenisation hazard worth stating. Re-encoding a prompt and its logged completion by concatenating *strings* lets byte-pair merges cross the boundary; concatenating token *ids* fixes it. The bug is renderer-dependent: on 50 probe items it changes the boundary index on 34/50 under the plain wrapper and 0/50 under a chat template, so it bites base checkpoints specifically. Relatedly, Qwen3 base tokenizers ship a chat template despite never having been tuned to follow one; automatic template selection dropped axis-E reproduction cosine to 0.13, and forcing the plain wrapper on base models restored all six archived checkpoints to ≥ 0.99992 .

4 Method

Five instruments, each pre-registered with a sha256 stamp before any statistic existed, with every deviation logged with its trigger and the data state at the time.

4.1 Instrument 1: the scale-panel test of the score that won

The reviewer of our previous draft identified the decisive scoping error: the 52-member panel had been spent replicating the score that *lost* the discrimination matrix, leaving the score that won — the logit-gap harmful margin, the only one whose confidence interval excluded zero at both aggregation units — untested at $n_{\text{lineage}} = 28$. This instrument runs it.

Four outcomes were pre-registered, verbatim, before any correlation: **HOLDS** requires $\rho \geq 0.50$ **and** a CI excluding zero at *both* units; **HOLDS_AT_MEMBER_UNIT_ONLY** is pre-committed as *not a win* — “this is the same unit-dependence iteration 4 documented and must not be written as one”; **COLLAPSES** converts the paper’s claim into a general statement about the score class; **REPLAY FAILED** halts the analysis entirely. The gate order is enforced by the driver: byte-identity of 17 reused library files, 14 offline apparatus assertions, constant extraction by **ast** (the iteration-3 driver calls **setrlimit** at module scope and cannot be imported), panel and ground-truth identity, a T0-REPLAY reproducing iteration 3’s 0.6673 [0.439, 0.904] / 0.929 to four decimals, and only then the pre-registration stamp. Ground truth is the archived judged plain-harmful refusal rate, reused rather than re-judged, so LLM spend is \$0.00 and the outcome variable cannot drift.

Three secondary analyses were registered in advance rather than chosen after: the archived-19 versus new-33 block split, which is the diagnostic that localised the paraphrase refit’s failure; a partial Spearman controlling for $\log_{10}$ parameter count, because the obvious rival explanation is that any activation score is a capability proxy; and both leave-one-lineage-out and leave-one-family-out jackknives.

4.2 Instrument 2: the paraphrase refit at scale

The AMS paraphrase refit is rerun on the identical 52 members, with four pre-registered outcomes: **R1** $\Delta_A > 0$ with its paired lineage-bootstrap CI excluding zero; **R2** $\rho(\text{refit A}) \geq 0.40$ with its CI excluding zero; **R3** $\Delta_B > 0$ where SET B is an *independently authored* paraphrase set; **R4** permutation $p < 0.05$ and off the floor by an order of magnitude. SET B was generated by a model that is never the judge, at temperature 0.3, and verified by the *frozen* iteration-3 **check_pair()** with zero hand-written repairs (80/80 strings pass, 78 on the first attempt); measured content-token Jaccard against SET A is 0.201.

4.3 Instrument 3: the read-versus-act re-analysis

A pure re-analysis of the frozen 30-checkpoint tree, with estimators imported rather than retyped and a 169-leg reproduction gate that passes at 10^{-6} , with the pooled coefficient and its interval reproducing at the archived seed to 0.0. Three repairs are made.

The coupling statistic. The primary estimand becomes the within-axis-A, across-member correlation over the 14 detection-powered members, at both aggregation units, with the lineage-clustered bootstrap and an exhaustive $7! = 5040$ permutation. The pooled 70-pair figure is retained only as a labelled secondary and is decomposed: because the design is a balanced 14×5 , a two-way decomposition of the rank cross-product is *exact and orthogonal*, so the share attributable to axis type is a measurement rather than an argument. A control ladder (drop the two by-construction null axes), a rank-residualised partial correlation, and a mixed-effects slope on ranks are all reported.

The verdict rule. Rather than assert that the rule is asymmetric, we simulate the study’s own prompt-clustered percentile bootstrap over a 141-cell grid of true AUROC $\times$ items per class, 2,000 replicates per cell with 2,000 inner resamples, and report the attainability surface of each verdict.

A closed-form Hanley–McNeil check accompanies it. The tally is then reported twice: as shipped over all 30 members, and restricted to the 14 members the pre-registration says the statistic exists on.

The ablated arm. The structural claim is re-carried on spontaneous refusal *rates*, which involve no AUROC: a two-sided Mann–Whitney U on member-level rates, an exhaustive permutation over all 293,930 group assignments (the arms share one tied rate, so `scipy`’s exact method is invalid here and its value is recorded but never quoted), a lineage-clustered bootstrap of the median difference, and a paired within-lineage sign test.

4.4 Instrument 4: breaking the label–axis lexical circularity, and the leakage control

Axis A is fitted on canned refusals and the detection label is a canned-refusal regex, so part of any shared AUROC is definitional. Instrument 4 re-labels 660 stratified items — stratified by regex label $\times$ stratum $\times$ projection tertile, with the middle tertile double-sampled and inverse-probability weights back to the item population — using the five-class semantic rubric already built for the degeneracy adjudication, loaded verbatim from the archived judge stage. A reproduction gate regenerates 667 archived cells from stored projections at $\max |\Delta| = 0.0$ first.

The same instrument runs the one published leakage control [Mehta, 2026]: four normalisation protocols on identical items and axes — the archived whole-pool centring, fold-internal centring under leave-one-prompt-out, fold-internal centre-and-scale (Mehta’s full residualisation), and a deliberately leaky whole-pool z-score — on axes A, B and the norm-matched random axis D, under both label sets. Axis D is the control on the control: a normalisation artifact would move it too. The leakage precondition is re-asserted rather than inherited, with axis-fit strings re-parsed from the archived fitting code and exact text overlap recomputed per member.

4.5 Instrument 5: the aggregation-unit repair, the threshold surface, and the claim ledger

A pure re-analysis over the frozen archives with no GPU and no spend ¹⁰: an 11-leg reproduction gate, every score recomputed at both aggregation units with the exhaustive permutation held constant, and a 164,736-point full factorial in the falsification battery’s five thresholds.

A second instrument audits the paper’s own prose ¹¹. Every numeric and verdict-string claim on every number-bearing surface — prose, tables, figure captions, figure summaries, abstract — is resolved against a two-tier pointer index over the shipped JSON (an unfiltered index over 152,118 numeric leaves resolves almost any two-decimal number to *something*, producing false matches, so only 51,178 reportable summary-statistic pointers may resolve a claim). 911 claims over 142 surfaces were audited; the regenerated text re-audits at 150 claims with zero flags, byte-identically across two runs, under a lint forbidding bare numerals and a mutation test confirming the pointers are live.

¹⁰Code: <https://github.com/AMGrobelsnik/ai-invention-c9546e-rating-model-safety-in-eighty-forward-pa/tree/main/round-4/evaluation-1>

¹¹Code: <https://github.com/AMGrobelsnik/ai-invention-c9546e-rating-model-safety-in-eighty-forward-pa/tree/main/round-5/evaluation-3>

5 Results

5.1 A cheap score that survives a fourfold increase in panel size

The pre-registered verdict is **HOLDS**, and it is the first positive result in this project. On 52 members over 28 lineages and 11 families, the first-decoding-step logit-gap harmful margin predicts the judged plain-harmful refusal rate at $\rho = 0.694$ [0.495, 0.822] at the member unit with a lineage-clustered bootstrap over 10,000 replicates, and $\rho = 0.564$ [0.140, 0.826] at the lineage-aggregated unit. Both criteria of the pre-registered rule — $\rho \geq 0.50$ and a CI excluding zero — are satisfied at both units. The Monte-Carlo lineage-permutation p sits at the design floor of 5×10^{-6} over 200,000 draws, and we quote the floor beside it rather than a smaller number the design cannot express.

Which cheap score predicts behaviour at 28 lineages

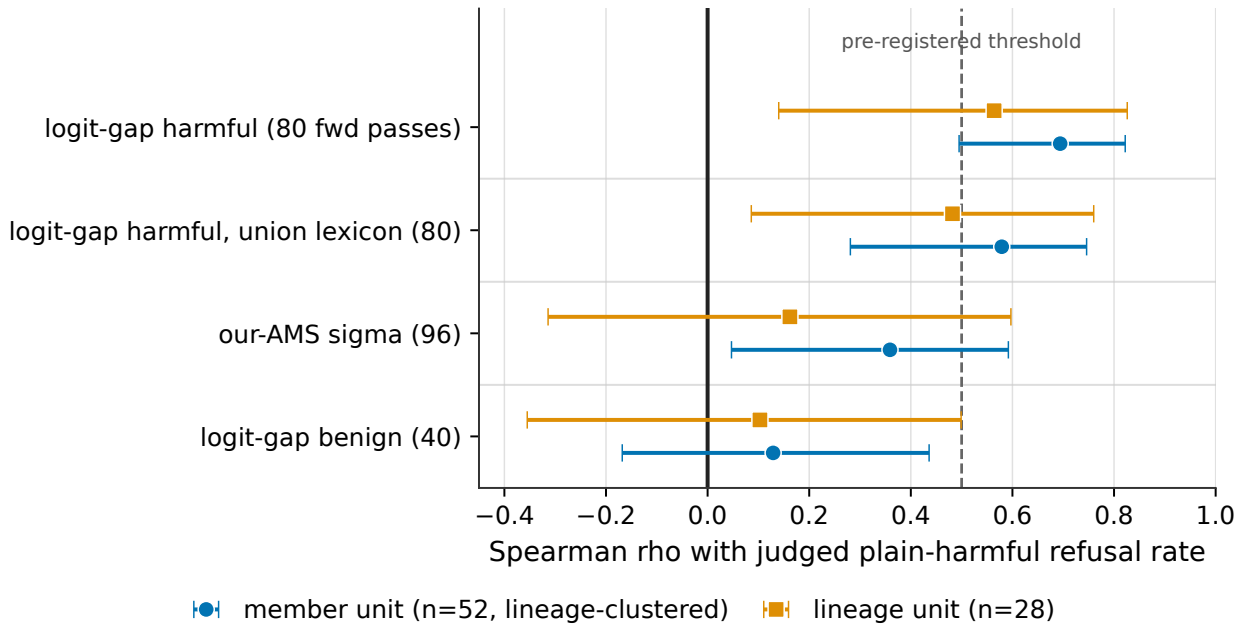


Figure 2: Spearman correlation of four benchmark-free scores with the judged plain-harmful refusal rate on the 52-member / 28-lineage / 11-family scale panel, at both aggregation units. Bars are 95% lineage-clustered bootstrap intervals over 10,000 replicates. Only the logit-gap harmful margin clears zero at both units; the benign-prompt variant of the same statistic does not clear zero at either.

Four things make this more than a large correlation.

It is not the small panel again. The decisive diagnostic is the one that localised the paraphrase refit’s failure, pre-registered here before any correlation: split the panel by provenance and ask whether the effect lives only in the block that produced it. It does not. The 19 archived members give $\rho = 0.6673$ and the 33 newly measured members $\rho = 0.6677$, a paired difference of -0.0004 $[-0.308, 0.380]$. The score transfers intact to 21 lineages it was never developed on.

It is not parameter count. The partial Spearman controlling for $\log_{10}$ parameter count is 0.676 [0.475, 0.814], essentially unchanged, and the score’s own correlation with $\log_{10}$ parameters is 0.092. The contrast with our-AMS σ is instructive: that score correlates with size at 0.458 and its partial drops from 0.359 to 0.243 $[-0.096, 0.507]$, an interval that covers zero. A reader entitled to suspect

score	fwd	gen.	ρ member [95% CI]	ρ lineage [95% CI]	perm p (floor)	LOLO	LOFO	AUC
logit-gap harmful	80	0	0.694 [0.495, 0.822]	0.564 [0.140, 0.826]	5.0×10^{-6} (5.0×10^{-6})	[0.661, 0.726]	[0.650, 0.772]	0.806
union lexicon	80	0	0.579 [0.281, 0.746]	0.482 [0.086, 0.760]	4.0×10^{-5}	[0.532, 0.611]	[0.520, 0.724]	0.750
our-AMS σ	96	0	0.359 [0.047, 0.592]	0.162 [-0.314, 0.597]	0.00988	[0.289, 0.389]	[0.289, 0.467]	0.534
logit-gap benign	40	0	0.129 [-0.168, 0.436]	0.103 [-0.355, 0.499]	0.43631	[0.009, 0.184]	[0.089, 0.177]	0.654

Table 1: The scale-panel result: four benchmark-free scores against the judged plain-harmful refusal rate on 52 members / 28 lineages / 11 families, at both aggregation units. All ρ are oriented (higher = safer) using the orientation map extracted from the iteration-3 driver. LOLO = leave-one-lineage-out (28 folds); LOFO = leave-one-family-out (11 folds); both are sign-stable in every fold for every score.

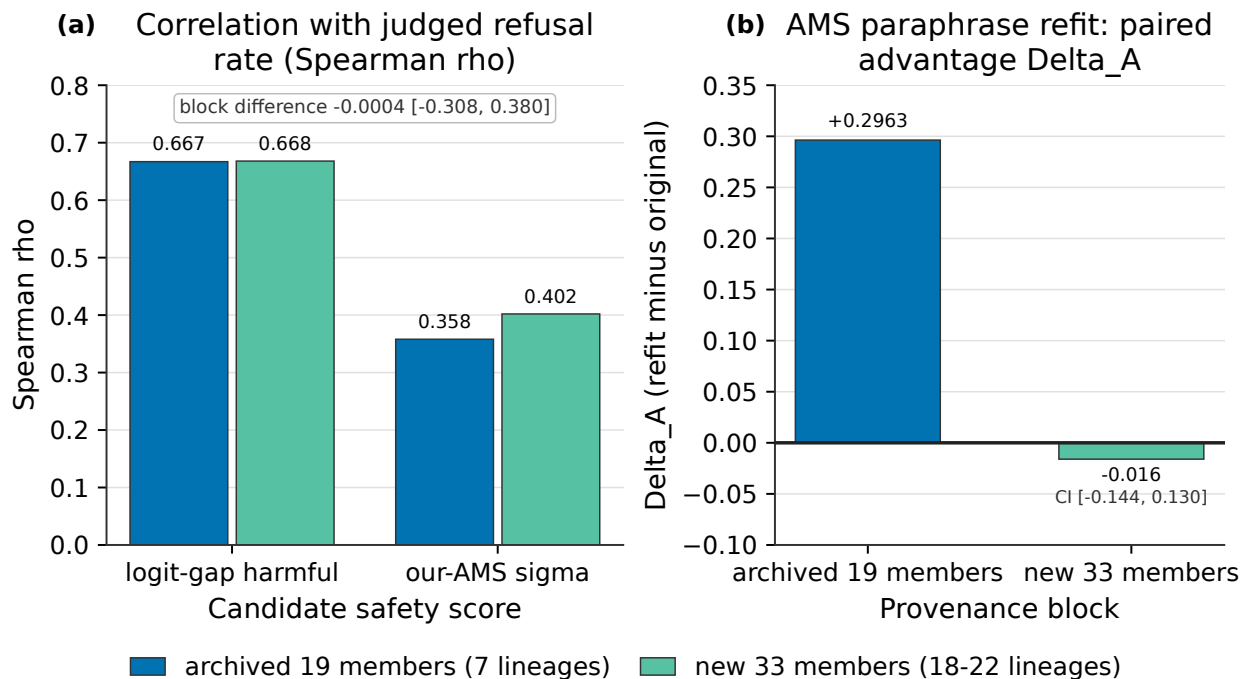


Figure 3: The provenance block split, the diagnostic that separates a transferable score from a small-panel artifact. The logit-gap harmful margin gives 0.667 on the 19 archived members and 0.668 on the 33 newly measured ones, a difference of -0.0004 . The AMS paraphrase refit’s advantage, by contrast, is entirely inside the block that produced it: $+0.2963$ archived against -0.016 new.

that any activation-derived score is a capability proxy gets a direct answer for both.

It is not one lineage or one family. Leave-one-lineage-out over 28 folds spans [0.661, 0.726] and leave-one-family-out over 11 folds spans [0.650, 0.772], sign-stable throughout. For comparison, the incumbent α_{50} ’s jackknife on the seven-lineage panel ranged from -0.086 to 0.771 depending on which single lineage was dropped.

It beats the anchor. Paired on the same resampled lineages, logit-gap harmful minus our-AMS σ is $+0.421$ [0.169, 0.684] with $P(\Delta > 0) = 1.000$, verdict `SCORE_BETTER`. Our-AMS σ itself reproduces the iteration-4 archived value on 49 of 52 members (maximum $|\Delta| = 0.0275$), and recomputing the 19 archived members from the models leaves the ranks *identical* (Spearman between iteration-3 and iteration-5 margins = 1.000, zero rank positions moved), so no Spearman statistic in this section is affected by the small numeric drift on three Llama members.

The harmful regime is load-bearing, which is why the product claim is narrowed rather than advertised. The benign-prompt variant of the identical statistic collapses to $\rho = 0.129$ [$-0.168, 0.436$], and the paired harmful-versus-benign difference is $+0.565$ [0.205, 0.873]. The score therefore reads the margin *on harmful prompts*. It is not harmful-prompt-free. What it removes is everything downstream of the forward pass — no generation, no judge, no benchmark, no reference model — and that sentence ships verbatim in the artifact rather than being written for the paper. The audit cost is 80 forward passes and 0 generations per checkpoint, with a median of 20.0s (p90 36.7s, maximum 70.1s) per member on one RTX A4500 *including model download*; the forward-pass count is the hardware-independent figure and the seconds are not.

Three plan assumptions were measured false and are recorded as deviations rather than quietly repaired. The five `UNRELIABLE`-flagged members the plan instructed us to exclude *do not exist* anywhere in the iteration-4 archive — neither the per-member table nor any per-member JSON carries such a field, and the string appears only inside verdict prose — so that exclusion set was not invented, and the with/without sensitivity was replaced by the block split and the missing-lexicon sensitivity, which are measurable. 51 of 52 rows carry a pinned revision SHA, not 52. And five members have no empirical refusal-onset lexicon for their tokenizer family; their primary columns are NULL, never back-filled, with the union-of-all-families secondary column ($\rho = 0.579$) reported beside them. Dropping those five leaves the primary ρ at 0.694 exactly, since they never entered it.

What we do and do not claim. We do not claim the scalar: the first-decoding-step refusal-affirmation margin is Li and Liu’s [Li and Liu, 2025], who establish it as a per-prompt safety margin and validate it against their own attack’s success rate across suffix strategies, not across models. We do not claim to be first on ablated checkpoints [Llorente-Saguer, 2026], nor the largest checkpoint panel [Hurtado, 2026]. What is ours is the conjunction none of them has: a model-level criterion validation of a first-decoding-step margin against a *graded* judged refusal rate, with no attested reference model and no weight access, over 11 architecture families, with inference clustered on the weight lineage — which zero of the five located model-level scores currently use. The closest published attempt at exactly this validation, AMS, reports Pearson $r = -0.546$ ($p = 0.043$) over 14 configurations from 4 families, and its directly comparable rank statistic is $\rho = -0.423$ at $p = 0.13$.

5.2 The companion negative: the paraphrase refit does not survive the same panel

The value of the previous subsection depends on a second score having been run through the identical apparatus and having failed, and one was. Our previous draft’s one forward-looking result was that refitting AMS’s contrast set on token-disjoint paraphrases lifted its correlation with judged behaviour from $\rho = 0.358$ to 0.654 on 19 members over 7 lineages. On the same 52 members over 28 lineages it does not replicate.

At the member level the original scanner reaches $\rho = 0.359$ [0.047, 0.592], the SET A refit 0.458 [0.197, 0.646], and the independently authored SET B refit 0.207 [−0.110, 0.463]. The paired advantage is $\Delta_A = +0.099$ [−0.027, 0.244] with $P(\Delta_A > 0) = 0.935$, short of the pre-registered interval criterion, so **R1 fails**. $\Delta_B = -0.152$ [−0.488, 0.075]: independently authored wording does not merely fail to reproduce the gain, it points the other way, so **R3 fails**. The permutation p is 0.135 against a floor of 5×10^{-6} , so **R4 fails**, and the 1/5040 floor the original result sat exactly on is retired by the larger panel. Only **R2** passes. The verdict is DOES_NOT_SURVIVE, with no salvage and no post-hoc subgroup.

The location of the failure is what makes it useful. The archived 19-member block reproduces $\Delta_A = +0.2963$ — a gap of 2.6×10^{-4} to the previously published +0.296, confirming the reuse is byte-exact rather than merely similar — while the 33 newly measured members give −0.016 [−0.144, 0.130]. Per block the correlation goes $0.358 \rightarrow 0.654$ on the archive and $0.402 \rightarrow 0.386$ on the new members. This is not a single-outlier story: leave-one-lineage-out over 28 folds keeps the shrunk Δ_A in [0.068, 0.122] and leave-one-family-out over 11 folds in [0.060, 0.137], never flipping sign. Three alternative calibration rules give +0.066, +0.152 and −0.035, none rejecting after Holm. The refit still *moves* AMS’s PASS/WARN/CRIT verdict class on $12/52 = 0.231$ [0.137, 0.361] of members; it just does not move them toward the truth.

score	archived-19 ρ	new-33 ρ	block difference [95% CI]
logit-gap harmful	0.667 [0.439, 0.904]	0.668 [0.365, 0.851]	−0.0004 [−0.308, 0.380]
our-AMS σ	0.358 [−0.072, 0.709]	0.402 [−0.048, 0.679]	−0.044 [−0.557, 0.514]
AMS paraphrase refit, Δ_A	+0.2963	−0.016 [−0.144, 0.130]	—

Table 2: The provenance block split, the diagnostic that separates a transferable score from a small-panel artifact. Two scores of nearly identical cost, one panel, opposite outcomes. The refit’s entire advantage lives in the 19 members that produced it; the logit-gap margin’s does not.

Read together, §5.1 and §5.2 say something neither says alone. It is not that cheap activation-derived scores are all illusory, which is the conclusion a reader would have drawn from the previous draft; nor that they are all fine. It is that at seven lineages the two are indistinguishable, and at twenty-eight they are not. That is the practical content of the small-sample-instability warning [Wang et al., 2026], stated in the one currency a practitioner cares about: which of two same-cost scores you would ship.

5.3 Reading versus steering: the coupling re-estimated, the verdict rule audited, the label broken

Our previous draft led §5 with the claim that reading and steering along one refusal axis are positively coupled at $\rho = 0.629$ [0.465, 0.803] over 70 (member, axis) pairs. The reviewer’s objection was that the 70 pairs are 14 members $\times$ 5 axes, that axis A is strong in both roles *by construction*

and axes C and D null in both roles *by construction*, and that the pooled Spearman therefore measures a between-axis-type contrast. Re-analysis confirms the objection and quantifies it.

The primary statistic is now the within-axis, across-member one, and it does not resolve. Within axis A across the 14 detection-powered members, $\rho = 0.547$ with a lineage-clustered 95% CI of $[-0.031, 0.930]$ over 7 resampling units and an exhaustive 5040-permutation $p = 0.149$ against an attainable floor of 1.98×10^{-4} . Aggregating members within lineage first leaves the sign unchanged at $\rho = 0.821$ $[0.348, 1.000]$. The defensible sentence is therefore: *the axis that induces is also the axis that reads, but among models the two qualities are only weakly and non-significantly related*. That remains a clean reversal of the induce-without-detect dissociation our earlier work claimed; it is not a demonstration of coupling strength, and the previous draft wrote it as one.

The confound is measured, not conceded. Because the design is a balanced 14×5 , a two-way decomposition of the pooled rank cross-product is exact and orthogonal. It attributes 0.896 of the pooled coefficient to between-axis-type variation, against 0.036 between members and 0.069 residual, the three shares summing to 1.000. Removing the axis main effect by rank-residualisation drops the association to $\rho = 0.234$ $[-0.059, 0.397]$; removing both the axis and the member main effects leaves 0.126 $[-0.240, 0.366]$; a mixed-effects slope on ranks gives 0.192 $[-0.075, 0.458]$. The trivial control the reviewer asked for is reported: dropping the two by-construction null axes moves the pooled coefficient from 0.629 to 0.545 $[0.284, 0.726]$ over 42 pairs. Within each single axis taken alone the coefficients are A 0.547, B 0.148, C 0.397, D -0.038 and E 0.416 — every one with an interval covering zero. No single axis carries a within-axis coupling on this panel.

The reviewer’s own recompute over thirteen members is reproduced exactly rather than paraphrased: dropping **Llama-3.2-3B-Instruct**, the one member whose axis-A verdict is **AMBIGUOUS** rather than **READS**, gives $\rho = 0.434$, $p = 0.14$, against this analysis’s 14-member $\rho = 0.547$, $p = 0.04$. Both of those p -values are the asymptotic Spearman value, which treats the 14 checkpoints as independent; the lineage-clustered interval covers zero at either n . The pre-registered verdict is **COUPLING_IS_AXIS_TYPE_CONTRAST** with **UNDERPOWERED** also firing — the within-axis interval’s half-width is 0.480, so at 7 lineages this panel could not have resolved a coupling of the size it estimates even if one is there. Both statements are true at once and the paper carries both. The within-member mean of 0.715 is demoted rather than defended: it is the mean of 14 coefficients computed over the *same* axis-type contrast on five points each, two of which are controls, so being larger than the pooled figure makes it weaker evidence, not stronger.

The verdict rule is n -asymmetric, and the Method previously misdescribed the gate. The reviewer observed that **READS** was issued at 7, 12, 28, 32 and 33 refusals — counts the artifact’s own **powered** column marks as not detection-powered — while only members with 0 or 1 refusals returned **UNDEFINED**, which is not the “fewer than 40 refusals” rule the Method described. Both halves are correct and are now fixed at the source. The code path is quoted in the deviation record **DEV-ITER5-01**: `verdict_from_ci` returns **UNDEFINED** if and only if the bootstrap CI bounds are non-finite, which happens because a ≥ 5 -per-class resample guard discards enough replicates when one class holds 0–1 items; `MIN_PER_CLASS = 40` governs a *separate* **powered** flag the verdict never consults. The corrected Method sentence appears in §4 above.

The asymmetry is then quantified rather than asserted, by simulating the study’s own prompt-clustered percentile bootstrap over 141 cells at 2,000 replicates each with 2,000 inner resamples. At

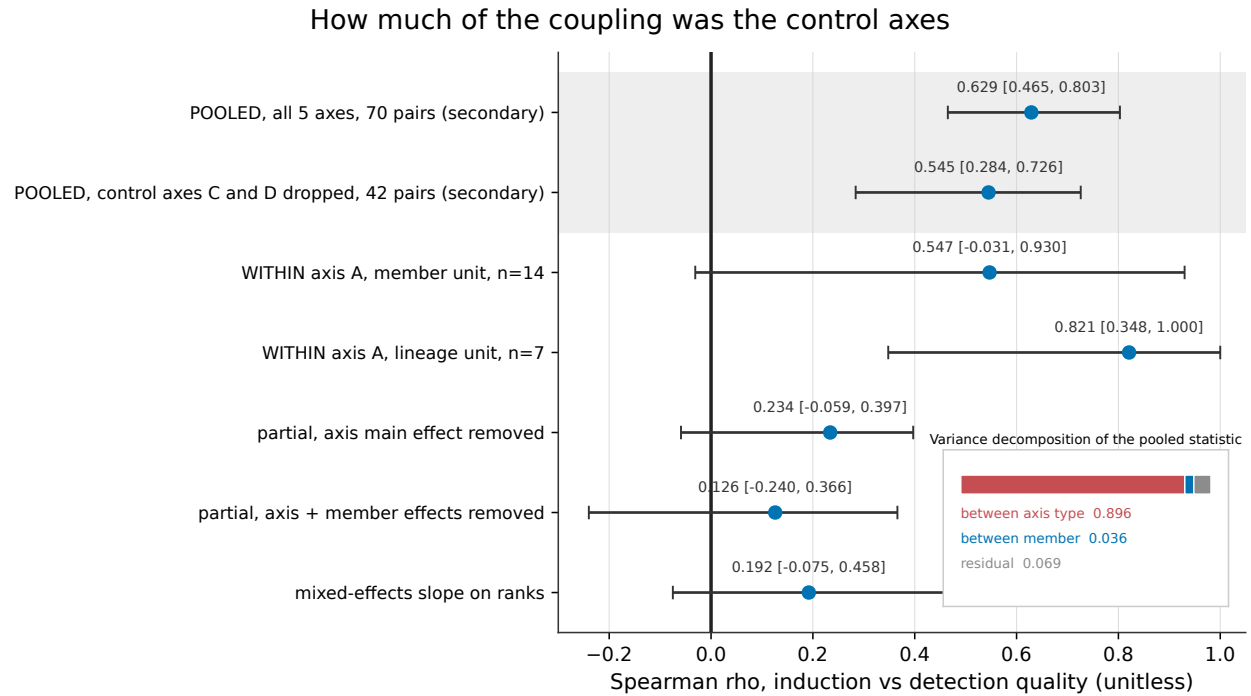


Figure 4: The read-versus-act coupling under seven estimators of the same relationship. The pooled 70-pair coefficient our previous draft led with (top) mixes between-axis and between-model variance; an exact two-way decomposition on the balanced 14×5 design attributes 0.896 of it to between-axis-type variation, 0.036 to between-member variation and 0.069 to residual. The primary estimand — within the canonical axis, across the 14 detection-powered members — has an interval covering zero at the member unit.

a true AUROC of 0.500, `AT_CHANCE` — which requires an entire 95% CI to fit inside the 0.20-wide band [0.40, 0.60] — is unreachable until $n = 80$ items per class: $P(\text{AT_CHANCE}) = 0.000$ at the pre-registered $n = 40$ gate, and the Hanley–McNeil closed form puts the i.i.d. threshold at $n = 65$. Under perfect separation `READS` fires with probability 1.000 at every one of $n = 7, 12, 28, 32, 33$. The asymmetry is *one-sided*, which matters for how the result should be read: the false-`READS` rate at true chance is only 0.005 at $n = 10$ and 0.001 at $n = 40$, so `READS` is not noise-driven — it is the *null* verdict that cannot be returned. Every “zero `AT_CHANCE`” sentence in this paper carries that footnote.

Accordingly the tally is reported twice. Over all 30 members the axis-A verdicts are 20 `READS`, 1 `AMBIGUOUS`, 0 `AT_CHANCE` and 9 `UNDEFINED`; restricted to the 14 detection-powered members they are 13, 1, 0 and 0. Reading is *measurable* — the AUROC and its interval both exist — on 21 members, not 20: the twenty `READS` members plus `Llama-3.2-3B-Instruct`. The minimum axis-A AUROC is 0.685 over the 21 members with a defined AUROC and 0.691 over the 20 `READS` members; the bare form “ ≥ 0.68 ” that appeared in our previous draft belongs to neither population and is retired.

The ablated arm no longer rests on underpowered AUROCs. The reviewer’s sharpest structural point was that the weight-edited arm’s five `READS` verdicts rest on counts of 12, 28, 32, 33 and 150, exactly one of which is powered. That is correct, and the claim does not need them. Abliteration removing the refusals to be read rather than the ability to read them is carried instead by spontaneous refusal *rates*, which involve no AUROC: a median of 0.0076 in the weight-edited arm and 0.0000 in the behavioural-uncensored candidate arm against 0.1131 in the aligned reference, over roughly 1,585 generations per member. A two-sided Mann–Whitney U separates the weight-edited arm from the aligned reference ($U = 13.5$, tie-corrected asymptotic $p = 0.0044$, 9 versus 12 members); because the arms share one tied rate, `scipy`’s exact method is invalid here and an exhaustive permutation over all 293,930 group assignments is reported in its place, giving $p = 0.0026$. A lineage-clustered bootstrap of the median difference over 9 lineages gives -0.1055 $[-0.2416, -0.0245]$, and over the 10 within-lineage ablated-versus-parent pairs the ablated member has the lower rate in 10 of 10 (exact paired sign test $p = 0.0020$, median paired difference -0.1669). The four underpowered AUROCs are cited as illustration only.

member	arm	n ref / com	spont. rate	powered	axis-A AUROC [95% CI]	verdict
<code>Qwen3-1.7B-Base</code>	aligned reference	146 / 146	0.1688	y	0.918 [0.871, 0.957]	<code>READS</code>
<code>Qwen3-1.7B</code>	aligned reference	197 / 197	0.2277	y	0.906 [0.859, 0.944]	<code>READS</code>
<code>Qwen2.5-1.5B-Instruct</code>	aligned reference	348 / 348	0.4023	y	0.763 [0.709, 0.812]	<code>READS</code>
<code>Llama-3.2-1B-Instruct</code>	aligned reference	172 / 172	0.1988	y	0.691 [0.603, 0.773]	<code>READS</code>
<code>Llama-3.2-3B-Instruct</code>	aligned reference	282 / 282	0.3260	y	0.685 [0.597, 0.763]	<code>AMBIGUOUS</code>
<code>Llama-3.2-3B-Instruct-abl.</code>	weight-edited ablation	150 / 150	0.1734	y	0.718 [0.628, 0.802]	<code>READS</code>
<code>Josiefied-Qwen3-4B-gabl.-v2</code>	weight-edited ablation	32 / 32	0.0202	N	0.998 [0.989, 1.000]	<code>READS</code>
<code>Josiefied-Qwen2.5-3B-abl.-v1</code>	weight-edited ablation	12 / 12	0.0076	N	0.889 [0.688, 1.000]	<code>READS</code>
<code>Qwen3-0.6B-abliterated</code>	weight-edited ablation	0 / 1572	0.0000	N	—	<code>UNDEFINED</code>
<code>Huihui-Qwen3-1.7B-abl.-v2</code>	weight-edited ablation	0 / 1574	0.0000	N	—	<code>UNDEFINED</code>
<code>TinyLlama-1.1B-Chat-v1.0</code>	aligned reference	7 / 7	0.0044	N	1.000 [1.000, 1.000]	<code>READS</code>
<code>Mia-001</code>	behavioural-uncensored cand.	0 / 1242	0.0000	N	—	<code>UNDEFINED</code>

Table 3: Twelve of the thirty per-member detection rows, chosen to show the whole operating range; the full 30-row table with norm-controlled readouts ships with the artifact. The two columns the previous draft omitted — refusal/compliance counts and the `powered` flag — are what let a reader see that `TinyLlama-1.1B-Chat-v1.0` reads at AUROC 1.000 on seven items and that the entire weight-edited arm has one powered member. Panel totals: 14 powered of 30; 20 `READS`, 1 `AMBIGUOUS`, 0 `AT_CHANCE`, 9 `UNDEFINED`.

The label and the axis share a lexical basis, and the AUROC mostly survives it. Axis A is the diff-in-means of hand-written canned refusals against canned compliances, and the label of record is an anchored regex over canned-refusal openers, so part of any shared AUROC is definitional. We measure it. Re-labelling 660 stratified items with the five-class semantic rubric moves the pooled axis-A AUROC from 0.834 [0.736, 0.923] under the regex label to 0.821 [0.752, 0.866] under the semantic one — a paired difference of -0.013 [$-0.067, +0.030$] at the member level and -0.024 [$-0.066, +0.018$] at the lineage level, with the two criteria agreeing at Cohen’s $\kappa = +0.789$ [$+0.699, +0.879$].

The deciding split is not the swap but the stratum. On canonically-worded refusals the axis separates at 0.897 [0.864, 0.922]; on the genuine refusals the regex misses (`REFUSAL_NONCANONICAL`) it reaches only 0.611 [0.542, 0.686], which does not clear these members’ own 20-draw random reading band, whose upper edge averages 0.750. The pre-registered verdict is `READS_CANONICAL_WORDING_ONLY` at both units. A caveat we did not expect and must carry: the rubric’s canonical/non-canonical split is *not* the regex’s split — 54 of 267 items (20.2%) that open with a frozen refusal opener are still called `REFUSAL_NONCANONICAL` by the judge, and the drift is member-dependent (0/27 on `Qwen3-1.7B-Base`, 17/25 on `Llama-3.2-3B-Instruct`). Taking the rubric class as “refusals the regex missed” over-counts 83 against 38. On the sharper subset — semantic refusal *and* regex non-refusal — the pre-registered floor of 40 is not met at $n = 38$, so the reportable claim is the pre-registered fallback: weighted corpus prevalence 0.0546 [0.0412, 0.0686], roughly one scored item in eighteen is a refusal the regex of record calls a compliance.

The one published leakage control, run on our own headline. Our thesis is that the item pool decides the result, and the single published control that tests exactly this had not been applied to the analysis it would test. It has now. Re-estimating every centring and scaling statistic inside the training fold under leave-one-prompt-out moves axis-A AUROC by -0.0205 [$-0.0352, -0.0071$] (centring alone: $+0.0009$; the deliberately leaky whole-pool z-score: -0.0205), and -0.0397 [$-0.0763, -0.0047$] under semantic labels — an order of magnitude short of the -0.336 the same control produced on its author’s own data [Mehta, 2026]. The control on the control holds: the identical protocol moves the norm-matched random axis D by only -0.0020 [$-0.0084, +0.0032$] and axis B by -0.0023 , so the axis-A movement is not pure normalisation. Zero fallback folds occurred anywhere, and text overlap between the scored items and the axis-fit strings is exactly zero on every member, re-asserted here rather than inherited. Verdict: `LEAKAGE_CONTROL_SMALL_DELTA`.

5.4 The canonical axis beats its paraphrase on semantics, not only on lexicon

An earlier draft of this work adjudicated rather than measured a semantic partial reversal: under a four-class judge the token-disjoint paraphrase axis B crossed a 0.50 refusal rate on every checkpoint, and we set that aside on the grounds that B’s high-coefficient text is degenerate. Filtering to text that passes the archived fluency screen *before* judging, and reporting every rate against a control floor measured on the same filtered population, replaces the argument with an estimate.

At matched axis-contrast units — axis A’s own 50%-refusal coefficient — the five-class any-refusal rate is 0.028 [0.008, 0.057] for axis B against 0.747 [0.618, 0.858] for axis A, with the false-positive floor at 0.146 set by the *random* axis D. The net quantity $B - \text{floor}$ is -0.118 [$-0.157, -0.082$] (paired prompt-clustered bootstrap, 5,000 replicates, $n = 600$ per axis): B sits below what a meaningless direction induces on the same population. The verdict is `REVERSAL_DOES_NOT_SURVIVE`, on 6 of 6 checkpoints and pooled.

Three sub-measurements make this an estimate rather than an argument, and each cuts against

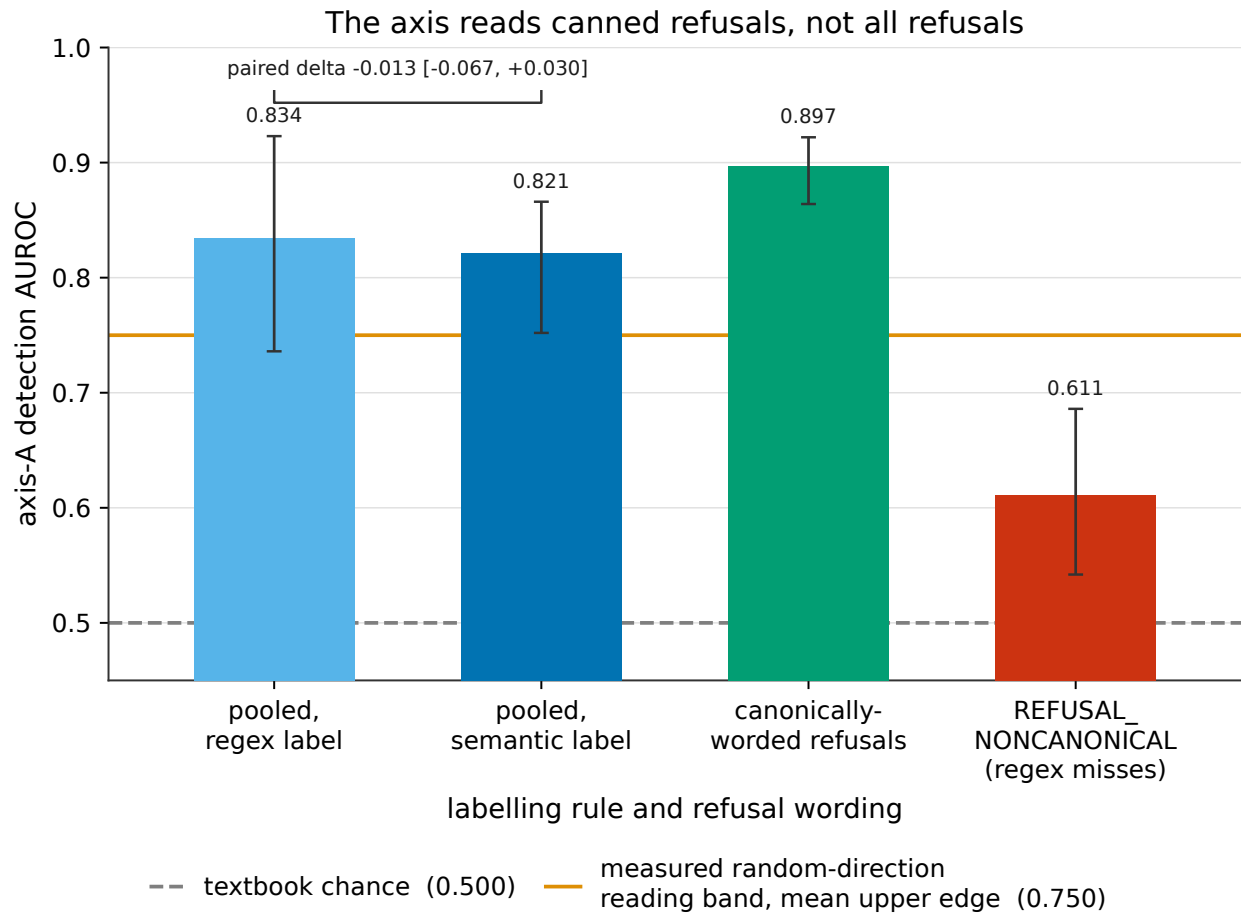


Figure 5: Axis-A detection AUROC under regex versus semantic labels, and split by refusal wording. Swapping the label barely moves the pooled value (-0.013 paired), but the axis separates canonically-worded refusals at 0.897 and the genuine refusals the regex misses at only 0.611 — inside the members’ own random-direction reading band, whose upper edge averages 0.750 . The verdict is `READS_CANONICAL_WORDING_ONLY`.

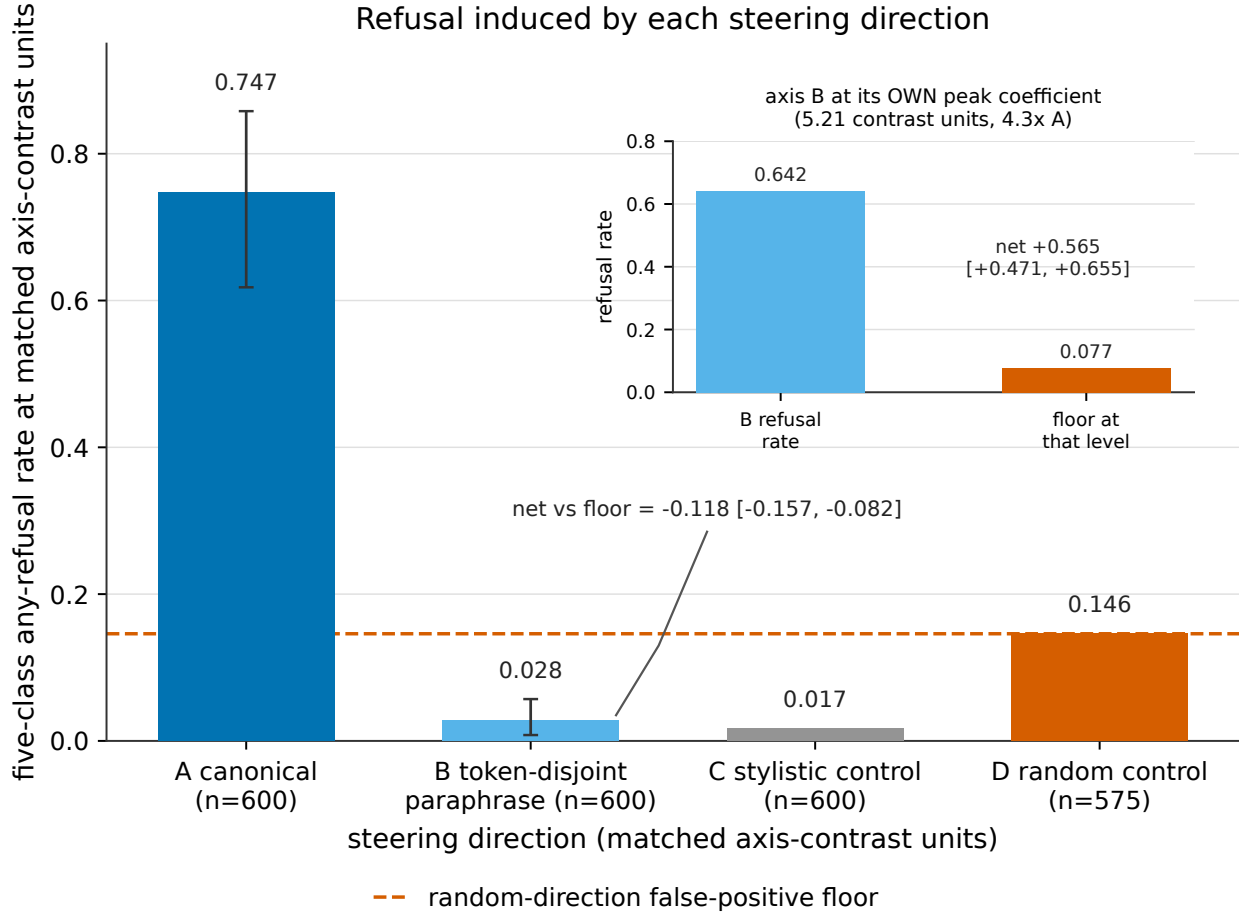


Figure 6: Five-class any-refusal rate on fluency-screened text at matched axis-contrast units, with the random-direction false-positive floor drawn as a reference line. The token-disjoint paraphrase axis B induces 0.028 against the canonical axis A’s 0.747 and sits 0.118 [0.082, 0.157] BELOW what a meaningless direction induces on the same filtered population, so the canonical axis’s advantage is semantic rather than a wording artifact. The inset shows the one pre-registered level at which B does clear its floor — its own peak coefficient, 4.3× the dose A requires.

something previously written. At matched contrast the lexical screen removes *nothing* — retention is 1.000 for both A and B — so B’s near-zero rate there is an absence of effect, not a filtering artifact, which inverts our earlier degeneracy story at the level that matters. At B’s own maximum coefficient (≈ 15 contrast units) retention does fall to 0.705, but 70.2% of the text that *passes* the screen is still judge-DEGENERATE, against 71.1% unfiltered: the lexical screen removes essentially none of the residual degeneracy, because the failure is semantic and the screen is lexical. And the control floor is itself made of screen-passing degenerate text — 59.0% of the random axis’s matched-cell survivors are judge-DEGENERATE — which is precisely why a semantic rate reported without a same-population floor is uninterpretable.

One pre-registered level splits the verdict, and it is the section’s nuance rather than a hedge. At B’s own peak refusal coefficient (5.21 contrast units, about $4.3\times$ the intervention A needs) B *does* clear the floor on fluent text: 0.642 against a floor of 0.077, net +0.565 [+0.471, +0.655], with only 4.9% DEGENERATE. B’s apparent reversal is real, but lives entirely at coefficients that matching forbids — which is what matching was introduced to detect. The Rogan–Gladen correction is reported throughout and is uninformative at the matched level by construction: both B’s rate and the floor fall below $1 - \text{specificity} = 0.196$, so both corrected prevalences truncate at zero and the corrected net is exactly 0 by construction rather than by measurement, which the artifact flags rather than quoting.

axis	n	anchored regex	four-class judge	five-class any	five-class non-can.	five-class degen.
A (canonical)	600	0.470	0.763	0.747	0.142	0.005
B (token-disjoint paraphrase)	600	0.002	0.043	0.028	0.018	0.002
C (stylistic control)	600	0.000	0.050	0.017	0.017	0.007
D (random control)	575	0.002	0.374	0.146	0.139	0.590

Table 4: Refusal rates on fluency-screened text at matched axis-contrast units, three scoring criteria side by side, pooled over the six depth-panel checkpoints. The random control’s four-class rate of 0.374 against its five-class degenerate rate of 0.590 is the false-positive floor any semantic steering claim has to clear. Criterion agreement is poor where it matters: Cohen’s κ between the regex and the five-class rubric is 0.424 on A, 0.108 on B and 0.020 on D.

Across all 30 read-versus-act checkpoints, matched contrast returns `NORM_MISMATCH_DOES_NOT_EXPLAIN` on 22, which rules out Petrov’s magnitude-collapse account [Petrov, 2026] on a panel five times the size of the previous test. On the two breadth-panel members that carried the objection that axis B does reach a 0.50 refusal rate, re-measurement at matched contrast finds 1 of 2 a genuine inducer (Llama-3.2-1B-Instruct) and 1 a norm artifact.

5.5 The aggregation unit, and a negative that is robust to its own thresholds

The most damaging defect a reader could have found in an earlier draft was internal: our AMS reimplementations’ correlation with judged behaviour appeared as 0.358 in one section and 0.821 in another, with a headline Δ computed from the second. Both numbers are correct and neither was labelled. At the **member level** — 19 checkpoints, resampled and permuted on the lineage label — the statistic is $\rho = 0.358 [-0.074, 0.699]$ with exhaustive permutation $p = 0.0911$. At the **lineage level** — 7 units, each the mean over that lineage’s defined members of both score and outcome — the same statistic is $\rho = 0.821$. The gap of 0.464 is what lineage aggregation buys by removing within-lineage variance and reducing n from 19 to 7.

That is not a bookkeeping repair, because the choice moves conclusions. Over the 16 score $\times$ configuration cells where both units are defined, changing nothing but the unit moves oriented ρ by

a median 0.238 and a maximum 0.557, and **flips the sign on 5**. The headline comparison inherits exactly that instability: on the carrier an earlier draft used, the oriented $\Delta = \rho(\alpha_{50}) - \rho(\text{our-AMS})$ is -0.929 $[-1.961, -0.113]$ at the lineage level and -0.376 $[-0.795, 0.110]$ at the member level — **SIGN_SURVIVES** but **EXCLUSION_LOST_AT_MEMBER_LEVEL**; on the discrimination matrix’s own carrier it gives -0.566 member and $+0.107$ lineage — **SIGN_FLIPS**, **EXCLUDES_AT_NEITHER**. The correct statement is that α_{50} loses to a cheaper activation scanner under every unit and carrier we can compute, and that no interval-based version of that claim survives both units. §5.1’s headline is reported at both units for the same reason, and it is one of the few rows in this study that clears zero at both.

score	member ρ	95% CI	lineage ρ	95% CI	CI excludes 0
α_{50} (max refusal rate), 19/7	−0.208	[−0.547, 0.175]	+0.321	[−0.887, 0.870]	neither
our-AMS σ , 19/7	0.358	[−0.074, 0.699]	0.214	[−0.765, 0.961]	neither
our-AMS σ , paraphrase refit, 19/7	0.654	[0.276, 0.859]	0.643	[−0.192, 1.000]	member only
logit-gap (benign), 19/7	0.101	[−0.243, 0.573]	0.286	[−1.000, 0.765]	neither
logit-gap (harmful), 19/7	0.667	[0.439, 0.904]	0.929	[0.412, 1.000]	both
our-AMS σ , scale panel 52/28	0.359	[0.047, 0.592]	0.162	[−0.314, 0.597]	member only
our-AMS refit A, scale panel 52/28	0.458	[0.197, 0.646]	0.224	[−0.229, 0.620]	member only
our-AMS refit B, scale panel 52/28	0.207	[−0.110, 0.463]	0.013	[−0.442, 0.453]	neither
logit-gap (benign), scale panel 52/28	0.129	[−0.168, 0.436]	0.103	[−0.355, 0.499]	neither
logit-gap (harmful), scale 52/28	0.694	[0.495, 0.822]	0.564	[0.140, 0.826]	both

Table 5: Every score against the judged plain-harmful refusal rate at **both** aggregation units. Rows 1–5 are the 19-member / 7-lineage panel with the exhaustive $7!$ permutation null in both units; rows 6–10 are the 52-member / 28-lineage scale panel with a Monte-Carlo null over 200,000 draws. The logit-gap harmful margin is the only score that excludes zero at both units on both panels.

With units named, the discrimination matrix stands unchanged and its negative is now robust to its own thresholds.

score	primary column	C1	C2	C3	C4	C5	pass	ρ (member)	95% CI	perm p	fwd/gen
α_{50}	max refusal rate	F	F	P	P	F	2/5	−0.208	[−0.545, 0.183]	0.3087	0 / 265
our-AMS σ	ams_sigma	F	F	P	P	F	2/5	0.358	[−0.072, 0.709]	0.0911	96 / 0
logit-gap (benign)	logit_gap_benign	F	F	F	F	F	0/5	0.101	[−0.243, 0.569]	0.6621	40 / 0
logit-gap (harmful)	logit_gap_harmful	F	F	F	P	F	1/5	0.667	[0.439, 0.904]	0.0038	80 / 0

Table 6: The discrimination matrix: four cheap benchmark-free safety scores $\times$ five falsification checks (C1 lexical, C2 monotone, C3 depth, C4 jackknife, C5 scorer; P = PASS, F = FAIL), on the frozen 19-member / 7-lineage panel, all correlations at the **member level** with the lineage as the resampling and permutation unit (exhaustive $7! = 5040$; achievable floor 1.98×10^{-4}). Verdict: **PROTOCOL_DOES_NOT_DISCRIMINATE**. Check 5 is a property of the shared scorer (REFUSAL $\kappa = 0.391$ against a 0.60 threshold), so it fails identically in every row and caps every row at 4/5.

The load-bearing observation is unchanged, and §5.1 sharpens it into something more uncomfortable than we could previously say: the score that predicts judged behaviour *best* passes the *fewest* checks, and it is now also the score that survives a fourfold panel increase while the score the checks favoured does not. Construct hygiene and predictive validity are close to orthogonal on this panel, and the checks pointed the wrong way. What is new is that the negative no longer depends on five arbitrary cutoffs. Over a 164,736-point full factorial in the five thresholds, **PROTOCOL_DOES_NOT_DISCRIMINATE** holds on a fraction 1.0000 of grid points, and on 0.9091 under

the stricter criterion that a rival must *strictly exceed* α_{50} ’s pass count. Exactly one single-axis change anywhere on the grid produces a strict rival win: lowering check 3’s depth-span threshold from 2.0 to 1.75. Check 5 cannot contribute at all, because its κ of 0.391 lies below the entire swept range $[0.40, 0.80]$ — proved structurally and verified empirically. Dropping the pass rules’ secondary clauses and scoring the numeric cutoffs alone lowers stability to 0.5802 and 0.2429, which locates the negative precisely: it is carried by the verdict-class and interiority clauses, not by the numbers.

5.6 Two empirical nulls that steering studies should adopt

The 30-checkpoint study was designed to test axes, and it produced two facts about *controls* that generalise beyond this paper.

First, a random direction is not behaviourally inert at the magnitude at which a refusal axis works. Injected at axis A’s own matched magnitude, a matched random unit direction induces refusal at a maximum rate of at least 0.10 on 7 of 30 members, with a worst case of 0.389 and a panel median of 0.028. Korznikov et al. [Korznikov et al., 2025] report the complementary effect — random steering raising harmful *compliance* to 1–13% at an identically calibrated coefficient — and never test random-induced refusal on benign prompts; this is that measurement, and it says the induction floor is a real quantity that steering claims must clear. Our own earlier random-direction null (0.00–0.058) was measured on six checkpoints and does not generalise to 30.

Second, a random direction does not *read* at 0.500. The empirical band of AUROCs over 20 random draws per member spans ± 0.075 to ± 0.500 across members, because residual streams are anisotropic, so a gate written against the textbook 0.500 is wrong by a wide and model-dependent margin. A single random draw is not a null distribution. Relatedly, a raw projection is $\|h\| \cos \theta$, so any direction inherits whatever refusal-versus-compliance *norm* difference the model has — one random axis “reads” at 0.171 on a member for that reason alone — which is why every AUROC in §5.3 is reported both raw and norm-controlled, and why the two agree to within 0.011 on the canonical axis.

6 Discussion

What a platform could do with this. The practical output of five iterations is one score and one protocol for trusting it. The score is 80 forward passes on a frozen 80-item harmful core, no generation, no judge, no benchmark, no reference model, roughly 20s per checkpoint including download at $\leq 4B$; it ranks checkpoints by judged harmful-refusal rate at $\rho \approx 0.69$ across 28 lineages and 11 families. It is a triage instrument, not a certificate: $\rho = 0.69$ leaves ample room for individual checkpoints to be mis-ranked, and the score reads harmful prompts, so an operator still has to hold 80 harmful strings — they just never generate from them, never send them to a judge, and never need a reference model. Against the honest alternative — a few hundred generations plus a judge per checkpoint — that is roughly three orders of magnitude cheaper per model and removes the one operational requirement (an attested reference) that the published ablation audit needs [Hurtado, 2026].

Three named pathologies, and the instances we claim. Three of this paper’s surviving results are measured instances of long-named methodological phenomena rather than new phenomena, and presenting them as discoveries would invite the correct objection. We name each and

claim only the instance.

Item-pool provenance is leakage. Kapoor and Narayanan’s taxonomy names our case directly as [L3.3] *sampling bias in test distribution* — “choosing a non-representative subset of the dataset for evaluation ... the test set is no longer representative of the general population about which claims are made” — with [L1.2] covering the statistics-estimated-on-everything half [Kapoor and Narayanan, 2023]. We claim the measured instance: on the six depth-panel checkpoints the *same* refusal axis moves from AUROC 0.486–0.790 when scored on an archived pool containing steered text to 0.906–0.980 when scored on each model’s own spontaneous text, with induction unchanged, so a read-versus-act conclusion was decided entirely by which items the score was evaluated on. The field-local precedent is exact: Mehta’s probe falls from 0.761 to 0.43 under leave-one-query-out with per-fold residualisation and scores 0.63 on a control condition where the effect cannot exist [Mehta, 2026].

The aggregation unit is ecological correlation. Robinson showed in 1950 that the same relationship computed at different levels of aggregation need not agree in sign: nativity against illiteracy is +.118 individually, −.526 over 48 states and −.619 over 9 divisions [Robinson, 1950]. Openshaw’s modifiable-area-unit literature separates the scale problem from the aggregation problem and reports that for a six-zone aggregation of 99 Iowa counties “the range of possible correlations is between −.99 and +.99” [Openshaw, 1984]. Simpson’s note [Simpson, 1951] names the paradox but its setting is categorical contingency tables, so Robinson is the closer analogue. We do not claim aggregation bias as a finding; we claim a measured instance in which it moves this study’s own headline by 0.464 and flips 5 of 16 signs.

The 7-to-28 collapse is small-sample correlation instability, and we are still below stability. Schönbrodt and Perugini define the point of stability and conclude that “in typical scenarios the sample size should approach 250 for stable estimates”; their Table 1 puts the critical n at 252, 238, 212 and 181 for true ρ of .1–.4 even at the most permissive corridor they report [Schönbrodt and Perugini, 2013, 2018]. We claim the instance against ourselves: quadrupling the panel from 7 to 28 lineages moves our own previously published Δ_A from +0.296 to +0.099 with an interval covering zero, localised so that the archived block reproduces to 2.6×10^{-4} while the new members give −0.016. And we state the uncomfortable corollary deliberately: $n = 28$ is still six to nine times below the point of stability for this effect-size band, so §5.1’s $\rho = 0.694$ is a direction of travel under a fourfold increase, not a settled value. The reference class this belongs to is small: of five published model-level internal safety scores, none resamples on the weight lineage and none exceeds four architecture families.

Why a validity battery can be right and useless at once. The battery’s cells each report a true property. What it cannot do is rank scores, because on this panel construct hygiene and predictive validity are close to orthogonal — the logit-gap harmful margin is at once the most predictive score, the least hygienic, and the only one that survives at 28 lineages. Had we ranked by the checks, we would have shipped the paraphrase refit and discarded the score that works. That is a stronger statement than the previous draft’s, which could only say the checks failed to discriminate; it now says they discriminated, in the wrong direction, on the one question a user cares about.

Limitations. (1) Scale: everything is measured at 0.13B–4.2B, and the within-family scale ladder runs only to 4B; nothing here licenses extrapolation to frontier checkpoints. (2) $n = 28$ lineages remains far below the point of stability (above), and the lineage-level interval on the headline,

[0.140, 0.826], is wide. (3) The score is not harmful-prompt-free, and the benign variant collapses to 0.129; any claim that this is a content-free audit is false. (4) The read-versus-act coupling is unresolved: 0.547 with a CI covering zero on 14 powered members from 7 lineages, and the members it is missing are systematically the ones with no refusals, so the estimate is conditioned on a model refusing sometimes. (5) The refusal axis reads canonically-worded refusals at 0.897 and non-canonical ones at 0.611, within the random reading band, so “the axis reads refusals” is true only of the wording it was fitted on. (6) Nothing here distinguishes an ablated checkpoint whose axis has been destroyed from one whose refusals have merely been suppressed, because the detection statistic requires refusals. (7) Our AMS reimplementation misses the published Table I by -6% , $+22\%$ and -40% on the three overlapping checkpoints, so every AMS comparison bounds *our reimplementation*; RAS and VISAGE were not run, for the checkpoint-overlap and cost reasons in §2. (8) Behavioural rates are judge-derived with a REFUSAL one-versus-rest annotator κ of 0.391, and our annotators are LLM agents, so every accuracy bounds agreement with an LLM panel rather than truth; disattenuated correlations are reported alongside raw ones, never instead. (9) Our fixed-lexicon maximum margin is a different estimand from Li and Liu’s per-prompt highest-logit affirmative token [Li and Liu, 2025], and five members’ tokenizer families have no empirical lexicon at all. (10) Everything steered is a statement about the steered dynamical system, which is provably not prompt-reachable [Mishra et al., 2026].

What we would do next. Three things follow, in cost order. Extend the panel toward the point of stability — the score costs 80 forward passes, so 250 lineages is a weekend of compute rather than a research programme, and it is the only way to convert a direction of travel into a value. Test whether the margin can be gamed: it is read at one token position from a fixed lexicon, which is exactly the surface a fine-tune could be tuned against, and the honest version of a triage score has to survive an adversary who knows it. And pair the refusal axis with a harm-intent axis [Llorente-Saguer, 2026] on the same ablated checkpoints, since their axis survives ablation to within 0.015 AUROC while the refusal channel goes silent; a two-axis signature — harm geometry intact, refusal channel dead — would be strictly more informative than either alone and would not need the attested reference the published ablation audit presumes [Hurtado, 2026].

7 Conclusion

We set out to build a safety score that costs seconds per checkpoint and touches no harmful text. The second half is not achievable by any route we found; the first half is. On 52 checkpoints over 28 weight lineages and 11 architecture families, a first-decoding-step logit-gap margin costing 80 forward passes, zero generations, zero judge calls and no reference model predicts the judged plain-harmful refusal rate at $\rho = 0.694$ [0.495, 0.822] at the member unit and 0.564 [0.140, 0.826] at the lineage unit, transfers between provenance blocks at 0.667 versus 0.668, survives a partial correlation on parameter count at 0.676, and beats our reimplementation of the closest published activation scanner by $+0.421$ [0.169, 0.684]. It reads harmful prompts, so what it saves is generation, judging, benchmarking and the reference model — not the harmful content itself.

The same panel retires the alternatives. The lexical-invariance refit of that scanner improves criterion validity by $+0.296$ on 7 lineages and by $+0.099$ $[-0.027, 0.244]$ on 28, with the archived block reproducing to 2.6×10^{-4} and the 33 new members contributing -0.016 . The act-side score this project began with, α_{50} , loses to a cheaper forward-pass scanner under every unit and carrier we can compute. The five-check battery built to explain that failure not only fails to rank scores

but ranks them backwards: it favoured the refit that collapsed and penalised the margin that held.

What is left is a score a platform could run tomorrow, and a measurement discipline that is the reason to believe it: three named pathologies — leakage through item-pool provenance, ecological correlation through the aggregation unit, and small-sample instability through panel size — each quantified on this project’s own previously published numbers, with the effect localised to the block that produced it. The reference class this score joins validates at four or fewer architecture families and never resamples on the weight lineage. On that evidence the useful claim is not that we found the right scalar; Li and Liu defined it. It is that the difference between a cheap safety score that works and one that does not is invisible at seven lineages and legible at twenty-eight — and that twenty-eight is still not enough.

A Corrections of Record

Twenty-six claims from earlier iterations are restated in the shipped artifacts rather than in the sections that first made them, each with the claim as previously stated, the corrected statement, the archived file and key it derives from, and why it moved ¹². The substantive items new to this iteration are: the read-versus-act coupling coefficient, demoted from a pooled 0.629 to a within-axis 0.547 with an interval covering zero, with the 0.896 between-axis-type variance share as the reason; the Method’s description of the UNDEFINED detection gate, which fires on a non-finite bootstrap interval at ≤ 1 refusal and not at the “fewer than 40 refusals” rule the Method stated, logged as deviation DEV-ITER5-01 with the three code paths quoted; the axis-A verdict tally, now reported twice (20/1/0/9 over 30 members; 13/1/0/0 over the 14 powered ones) with an attainability footnote establishing that AT_CHANCE is unreachable below $n = 80$ per class; the reading claim, narrowed to READS_CANONICAL_WORDING_ONLY (0.897 canonical versus 0.611 non-canonical against a random band edge of 0.750); the ablated-arm claim, re-carried on refusal rates rather than on four underpowered AUROCs; the minimum axis-A AUROC, which is 0.685 over the 21 members with a defined AUROC and 0.691 over the 20 READS members, with the bare “ ≥ 0.68 ” retired; the count of members on which reading is measurable, which is 21 and not 20; a stale top-line summary of 18 READS / 0 AT_CHANCE / 10 UNDEFINED, diagnosed exactly as an intermediate log state with the AMBIGUOUS class dropped, which is why it sums to 28; and reference [Basu et al., 2026], completed to its full eight-author list. Carried forward unchanged: the AMS paraphrase refit (DOES_NOT_SURVIVE); the semantic-reversal adjudication (REVERSAL_DOES_NOT_SURVIVE at matched contrast, REVERSAL_SURVIVES at B’s unmatched peak); the archived relative depth of 0.25; the random-direction null, rescoped to a measured induction floor reaching 0.389 on 30 checkpoints; the early-warning-signal arm, closed with a direction-specific difference-in-differences of -2.334 $[-3.573, -1.037]$ that fails Holm within its 48-test family and would need on the order of 1,880 prompts; the observable-validity gate, which admits 0 model pairs at the layer- L readout and 1 at the final-layer readout; the relaxation-rate claim, withdrawn as non-identifiable on 640 of 640 rows; the α_{50} accounting, where the primary logistic estimator is DEFINED on 1 of 19 members and that member is itself excluded; and nine bibliographic corrections against the arXiv API, including reference [Llorente-Saguer, 2026], whose title is *The Geometry of Harmful Intent*. Other work consulted in the course of these corrections and cited for completeness includes Rahimi et al. [2026], Kwon [2026], Taimeskhanov et al. [2026] and Allal et al. [2025].

¹²Code: <https://github.com/AMGrobelt/ai-invention-c9546e-rating-model-safety-in-eighty-forward-pa/tree/main/round-3/evaluation-2>

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